Experiment 19: Word Sense Disambiguation using Lesk in Python

1. Aim

To write a Python program for word sense disambiguation using the Lesk algorithm.

2. Procedure

1. Import `lesk` from `nltk.wsd` and `word_tokenize` from `nltk.tokenize`.
2. Define two sentences containing an ambiguous word in different contexts.
3. Tokenize the sentences and apply `lesk()` on the target word.
4. Retrieve the disambiguated WordNet synset and its definition.
5. Display the predicted sense and definition for both sentences.

3. Program

```
from nltk.wsd import lesk
from nltk.tokenize import word_tokenize

s1 = lesk(word_tokenize("I deposited money in the bank"), "bank")
print("Sense 1:", s1.name(), "-", s1.definition())

s2 = lesk(word_tokenize("The boat was tied to the river bank"), "bank")
print("Sense 2:", s2.name(), "-", s2.definition())
```

4. Output

```
Sense 1: savings_bank.n.02 - a container for keeping money safely
Sense 2: bank.n.01 - sloping land (especially the slope beside a body of
water)
```

5. Result

Thus, the Python program was successfully executed to perform word sense disambiguation using the `lesk()` algorithm.